Laser Interferometer (Michelson)

1. Definition

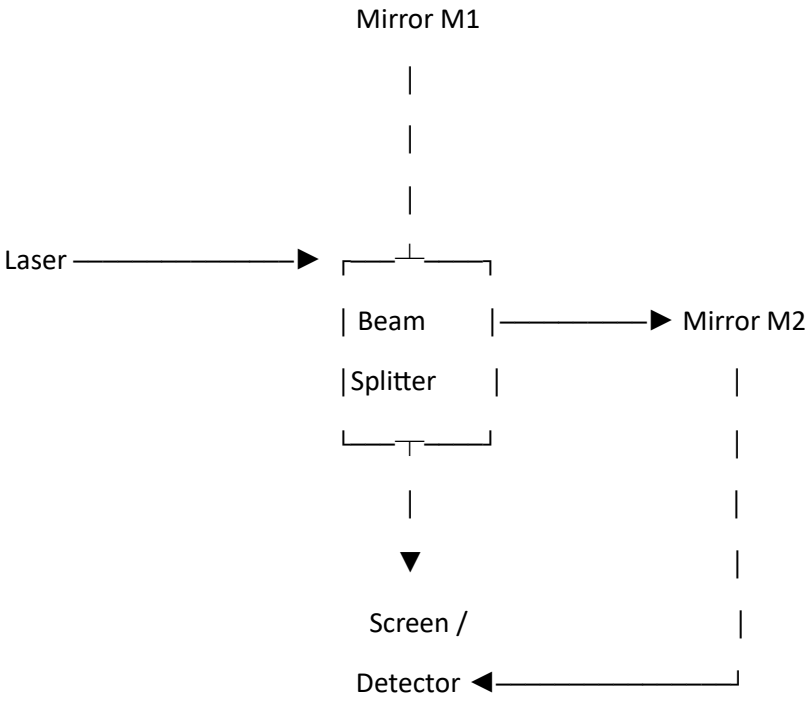
A Michelson interferometer is an optical instrument that uses the principle of interference of light to measure very small distances, changes in wavelength, and refractive index.

2. Principle

It works on the principle of superposition of light waves. A coherent laser beam is divided into two beams by a beam splitter. The beams travel along different paths, reflect from mirrors, and recombine to produce an interference pattern.

- Constructive interference: Bright fringes are formed.
- Destructive interference: Dark fringes are formed.

3. Experimental Setup and Diagram



The diagram is schematic; the beam splitter directs light along two perpendicular paths, which recombine to form interference fringes.

4. Components

1. Laser source: Provides coherent, nearly monochromatic light.
2. Beam splitter: Divides the incident beam into two beams.
3. Mirrors (M1 and M2): Reflect the beams back towards the beam splitter.
4. Screen or detector: Displays or measures the interference pattern.

5. Important Formulae

Optical path difference for the two beams:

$\Delta = 2d$

Here, d is the difference in the effective mirror-arm lengths, assuming the same medium and normal incidence.

Condition for constructive interference:

$\Delta = m\lambda$

Condition for destructive interference:

$\Delta = (m + 1/2)\lambda$

Where:

- Δ = optical path difference
- λ = wavelength of light
- m = integer (0, 1, 2, ...)

If a mirror moves by a distance x, the optical path difference changes by 2x.

If N fringes pass a reference point during this movement:

$2x = N\lambda$

Therefore:

$\lambda = 2x/N$

6. Procedure

1. Direct the laser beam towards the beam splitter.
2. Adjust the mirrors so that the two reflected beams overlap.
3. Observe the interference fringes on the screen or detector.
4. Move one mirror by a small, measured distance.
5. Count the number of fringes passing a fixed reference point.
6. Calculate the wavelength using $\lambda = 2x/N$, if wavelength measurement is the objective.

7. Sample Calculation

Suppose a mirror moves by 0.158 mm and 500 fringes pass the reference point.

Given:

$x = 0.158 \text{ mm} = 1.58 \times 10^{-4} \text{ m}$

$N = 500$

Using:

$\lambda = 2x/N$

$\lambda = (2 \times 1.58 \times 10^{-4})/500$

$\lambda = 6.32 \times 10^{-7}$ m

Therefore, the measured wavelength is 632 nm.

These are illustrative sample values, not actual experimental readings.

8. Applications

1. Measurement of the wavelength of light.
2. Measurement of extremely small displacements.
3. Determination of refractive index.
4. Testing optical surfaces and components.
5. Precision optical measurements.

9. Result

The Michelson interferometer demonstrates the interference of coherent light beams. Changes in optical path difference shift the fringe pattern, allowing precise measurements of wavelength and small displacements.